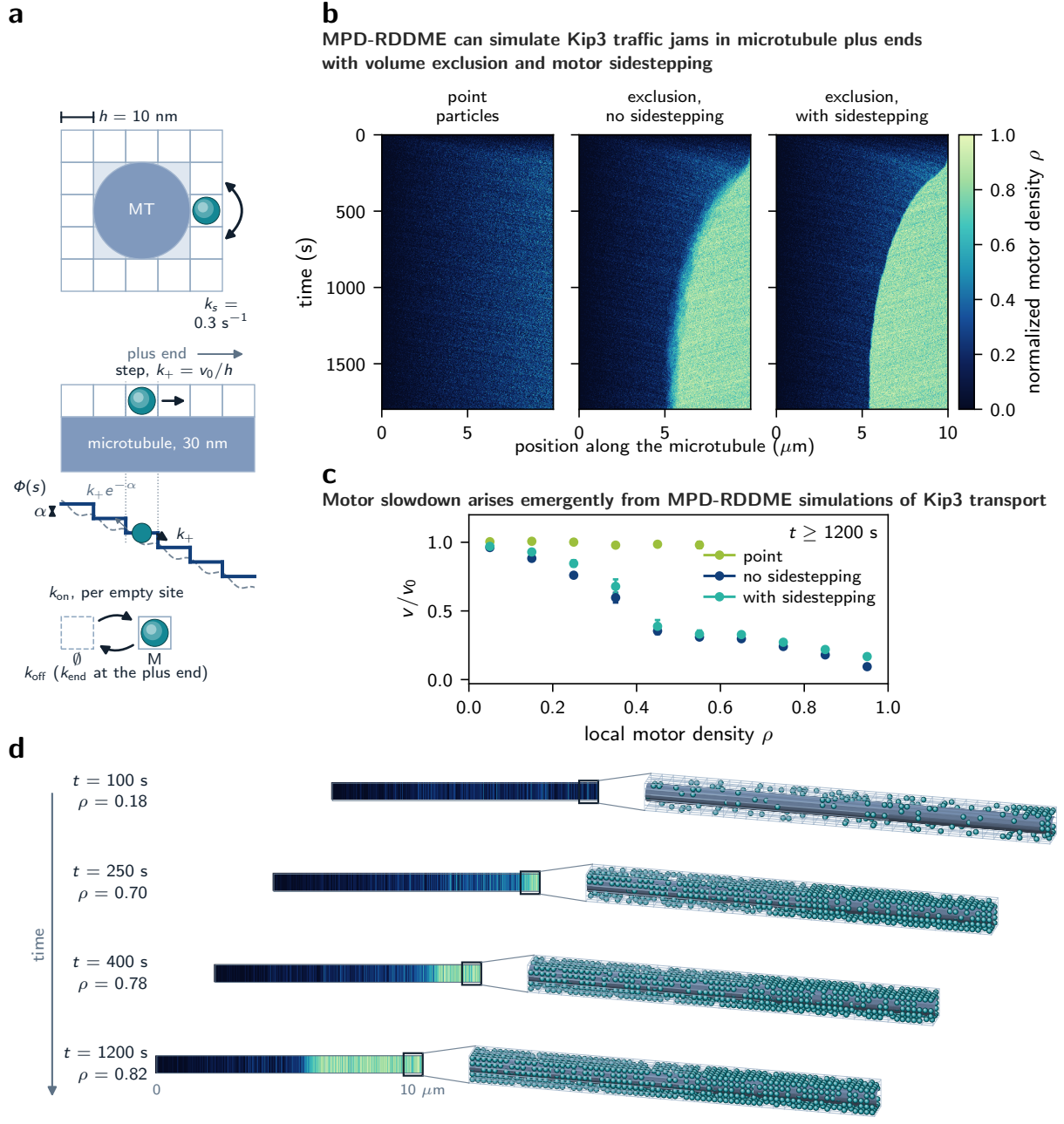

a
h = 10 nm
MT
k_s = 0.3 s^-1
plus end
step, k_+ = v_0/h
microtubule, 30 nm
Φ(s)
α
k_+ e^-α
k_+
k_on, per empty site
∅
M
k_off (k_end at the plus end)
b
MPD-RDDME can simulate Kip3 traffic jams in microtubule plus ends with volume exclusion and motor sidestepping
point particles
exclusion, no sidestepping
exclusion, with sidestepping
time (s)
0
500
1000
1500
position along the microtubule (μm)
normalized motor density ρ
c
Motor slowdown arises emergently from MPD-RDDME simulations of Kip3 transport
t ≥ 1200 s
point
no sidestepping
with sidestepping
v / v_0
local motor density ρ
d
t = 100 s
ρ = 0.18
t = 250 s
ρ = 0.70
t = 400 s
ρ = 0.78
t = 1200 s
ρ = 0.82
time
10 μm